

Comprehensive Analysis of the Effect of Social Influence and Brand Image on Purchase Intention

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Abstract

Consumers live in a social media environment and naturally form a set of trust mechanism from close circles, internet celebrities, and online articles. However, no sufficient research to examine these relationships with social influence and the effect of trust on social influence. This study integrates these factors and brand image to examines their impact on purchase intention. A questionnaire and SEM were used to test the reliability and validity of the hypothesis. The results support that purchase intention is significantly influenced by social influence and brand image. The outstanding contributions are as follows: first, it expands the research scope of social influence; second, it is found that close circles is the vital dimension of social influence, followed by social media viewership and e-WOM readership. Finally, based on the trust mechanism, marketing suggestions are put forward.

Plain Language Summary

Social Influence and Trust Mechanisms

Consumers live in a social media environment and naturally form a set of trust mechanism from close circles, internet celebrities, and online articles. However, no sufficient research to examine these relationships with social influence and the effect of trust on social influence. This study integrates these factors and brand image to examines their impact on purchase intention. A questionnaire and SEM were used to test the reliability and validity of the hypothesis. The results support that purchase intention is significantly influenced by social influence and brand image. The outstanding contributions are as follows: first, it expands the research scope of social influence; second, it is found that close circles is the vital dimension of social influence, followed by social media viewership and e-WOM readership. Finally, based on the trust mechanism, marketing suggestions are put forward.

Keywords

social influence, friendship, e-WOM, celebrity, trust, purchase intention

Introduction

Social influence has a profound impact on the general consumers' attitudes, beliefs, and actions (Argo & Dahl, 2020; Cambra-Fierro et al., 2021; Hsu & Lin, 2016; Kelman, 1958; Mi et al., 2019). This phenomenon is even emphasized in today's platform marketing strategy. It present an unprecedented breadth and depth of influence and most of company get much fetch from platform, and resulting in a marketing industry worth more than \$13 billion (Rahal, 2020), up to \$13.8 billion in 2022 (Influencer Marketing Hub report, 2021). About 92% of social network users said they had followed at least one

celebrity or influencer, 36.2% of users had tried products recommended by celebrities, and 20.4% of users had purchased products recommended by the influencer (Statista, 2021). About 90% of marketers believe that

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influencer marketing is effective (Dimitrieska & Efremova, 2021). It has been documented that these profits come from the influence and credibility of key opinion leader, micro-influencer, blog influencer, celebrity influencer, and social media influencer (Argo & Dahl, 2020; Dimitrieska & Efremova, 2021; Hsu & Lin, 2016; Mi et al., 2019). These social influence mechanisms have thoroughly changed the way of product promotion and communication with users, boosting product sales. Maximizing sales with the least cost is the most important thing for businesses. However, the research on the difference in the degree of influence of social influencers on consumer behavior change is lacking. Therefore, this study aims to provide a basis for enterprises to formulate marketing plans according to the influence of different celebrities.

Trustworthiness is netizen's perception of the influencers' integrity, expertise and competence and as the core antecedent factor affect consumer behavior (Argo & Dahl, 2020; Hsu & Lin, 2016; Mi et al., 2019). Trust in psychologically refers to a party's confidence in the trustee's positive behavior and expects better results and valuable resources from cooperation and as an essential prerequisites for building relationships between strangers. Trust transfer theory states that an individual's trust can be transferred from a trusted source to an unknown target if there is a specific association between them (Doney & Cannon, 1997). Thus, therefore, various types of influencers can influence the attitude, cognition, and behavior of Internet followers toward the products they endorse. Kirschman (2022) pointed out that the originality of the post, the number of followers, and the brand's name prominently placed can affect the credibility of the product, while simply publishing a new product post has no effect. Such as the famous soccer Cristiano Ronaldo, who has more than 500 million followers (Al Rabadi, 2022), and can be able to draw their attention to the brands he's promoting. But studies have noted that the more famous a football personality's value, the more positive and negative comments they garner (Kucharska, 2018). Not surprisingly, family and close friends and their recommendations are considered the most reliable and valuable (Dimitrieska & Efremova, 2021). Stemming from familiarity with each other's preferences, close friends can be seen as customized experts and give sound judgments (Gellerstedt & Arvemo, 2019); On the other hand, they can become good friends, which indicates that homogeneity is relatively high, which affects trustworthiness and expertise and therefore credibility (Ayeh et al., 2013; Gellerstedt & Arvemo, 2019). However, most extant studies have focus on credibility of influencers, while the consumer's psychological

mechanism (i.e., trust) changes has not been well explained. This may has lead to the neglect of other variables (psychological factors) in the process of transmitting trust.

To fill this gap, we introduce social influence and brand image in this study as key factors influencing trustee's trust transfer to brand purchase intention. Social influence is one of the most critical sources influencing consumers' behaviors actively or passively (Argo & Dahl, 2020; Cambra-Fierro et al., 2021). Brand image has been identified as one of key antecedent directly influencing purchase intentions (H. W. Kim et al., 2012; Leader et al., 2021; Lien et al., 2015; Y. H. Wang & Tsai, 2014).

Social influence in network is more diverse and complex than the traditional influence of friends and salespeople. For the research purpose, family and close friends, key opinion leaders, micro-influencers, blog influencers, celebrity influencers, and social media influencers can be classified as follows: face-to-face friendship, e-WOM readership, and social media viewership. "These types of communication are the most important sources of information about products or services in digital contexts" (Tobon & García-Madariaga, 2021, P607) and then examine they relationship with social influence, between social influence along with brand image and brand purchase intentions, meanwhile take trust as a moderating variable.

Literature Review and Hypotheses Development

Social Influence Theory

Social influence theory proposed by Kelman (1958), refers to an individual's attitudes, beliefs, and subsequent actions or behaviors influenced by referent others (Bartal et al., 2019), which include compliance, identification and internalization. Burger et al. (2001) pointed compliance occurs in one's response to the request of the person they like or approve based on reciprocity influence, and the more significant perceived similarity between two people will enhance compliance, even the apparent similarities (Burger et al., 2001; M. Kim & Kim, 2020). By the same logic, consumer comply with celebrities with physically attractive personalities that match consumers (Loughran Dommer et al., 2013; Zhuang et al., 2021). Identification usually occurs in these two scenarios: hope to gain approval or avoid punishment (Deutsch & Gerard, 1955).

And only trusted information has a real influence, which genuinely change personal attitude and behavior (Turner & Oakes, 1986). Thus, information source credibility is the prerequisite factor for influencing people's

behavior (Fleury et al., 2020; Schwarzwald et al., 2001; Turner & Oakes, 1986; T. Zhang et al., 2020). Mi et al. (2019) suggested that the potential source of influence is interpersonal and social; interpersonal influence comes from family and friends. Socially related forces refer to the power of online social networks (Zhuang et al., 2021). Social influence can be described using different constructs by different theories depending on the research context.

To get an in-depth study of social influence, we also compare those fields mentioned above to understand how trust establishes from the social influence circles and further impacts consumers' purchase intention.

The Social Influence and Reflective Indicator

Face-to-Face Friendship. Face-to-face friendship here refers to friends sharing common interests and thinking alike, familiar with each other's personalities and preferences (Gellerstedt & Arvemo, 2019). Friendship is a critical driver of personal life satisfaction and psychological well-being (M. Kim & Kim, 2020). Individuals follow their peer's suggestions because of the congruence of value systems, language similarity, and interest similarity (M. Kim & Kim, 2020), or fear of losing peer support (Engels et al., 2004), sometimes to build and maintain friendships (Tucker et al., 2014). In addition, personal attitudes and self-concepts can be influenced by family, friends, or colleagues (Yang et al., 2007).

Godey et al. (2016) showed that 81% of teenage girls get trend information and 45% of decision-making advice from their friends and family. Studies have proved that when shopping with friends, personal behavior changes: increases purchase amount, spend more money, positive attitudes toward merchants, and increased non-compulsive impulse shopping (De Vries et al., 2018).

Online Social Network. The occurrence of social influence does not require purposeful or conscious efforts to change the behavior of others, nor does it require direct contact between two actors. This kind of social influence is of particular interest in the subfield of social networks and online purchasing as it links the structure and composition of networks and the individual behaviors and outcomes of actors embedded in them (Filieri et al., 2018; Perry & Ciciurkaite, 2019). The direction of influence theoretically depends on the attitude, attribute, or behavior of the participants in the network (Bartal et al., 2019) and the relationship between participants in social networks and the extent of the influence (Argo & Dahl, 2020). Further expanding based on previous research on the forces of social influence, Zhuang et al. (2021) classified social influence as the influence from the information, action, and structure, which include duration, intimacy, emotional intensity, reciprocal services between

users, and interaction activity between people (type, frequency, and context). Based on the research results of Zhuang et al. (2021), we summarized social influence types within online influence as follows:

E-WOM Readership. Reviewer online (e-WOM readership) refers to any positive or negative statement about a product or company made by a potential, actual, or former customer that can be made available to the people over the Internet and is considered trustworthy (Alsoud et al., 2022; Chih et al., 2013). Social media and social platforms allow consumers to share their shopping and product usage experiences. The reliable and trustworthy source of information that provides referential and diagnostic for the potential buyer, who can overcome purchase uncertainty (Al-Adwan et al., 2022; Putri Utami et al., 2020), and avoid the punishment caused by making a wrong decision (Singh & Matsui, 2017; Bastos & Moore, 2021). Consumers are used to searching for unfamiliar product information online and trust other peer comments more than marketers (Al-Adwan et al., 2022; Doh & Hwang, 2009; Mi et al., 2019). And research indicated that the information from Online Consumer Reviews (OCR) is more useful than that from opinion leaders (OL), and the influence is more potent when OCR includes the shopping experience. Meanwhile, consumers will be less influenced by OLs as they have more online shopping experience (Tobon & García-Madariaga, 2021).

E-wom's credibility is related to the writer's expertise, relevance, timeliness, accuracy, comprehensiveness, and quantity of the content (Majali et al., 2022; Putri Utami et al., 2020; Sa'ait et al., 2016). The good or bad signals of products obtained by online searches may directly or indirectly influence personal online shopping decisions (Alsoud et al., 2022). Therefore, e-WOM occurs in choosing phone brands (Yang et al., 2007), tour destinations (M. Kim & Qu, 2017), and green products choice (Mi et al., 2019).

Social Media Viewership. Live streaming refers to marketers' promotion and sale of goods through influential mainstream media on social media channels alongside charismatic online celebrities. And live streaming e-commerce is developing rapidly, and the amount of merchandise sold via live streaming is staggering (Meng et al., 2021). Through their individuality, expertise, and attractiveness, persuasion-oriented performances, celebrities influence consumers' attitudes toward endorsed products (Huhn et al., 2018; Meng et al., 2021; Mi et al., 2019). Celebrity congruence with products or congruence with the consumer's ideal self-image are particularly persuasive (Choi & Rifon, 2012; Hui, 2017). People tend to reflect on their real self-image or express their ideal

self-image through the product (Mi et al., 2019). When people find that celebrities can promote their self-image or demonstrate congruence with their perfect self-image, they will adopt the recommended suggestions and even imitate the celebrity's behavior to change themselves (Loughran Dommer et al., 2013; Zhuang et al., 2021). Meanwhile, online celebrities who can arouse consumers' emotional trust will influence their decision (Meng et al., 2021). This is typical internalization behavior of social influence (Loughran Dommer et al., 2013). Another internalization study of social influence showed that a non-smoker in a network became a smoker because the person's network included a smoker; this is the most robust evidentiary material (Simons-Morton & Farhat, 2010).

Based on summarizing the literature mentioned above, it is found that consumers' behavior and decision-making are significantly affected by face-to-face friendship, e-WOM readership, or social media viewership, which have core features: trust the influencer and get the benefit. As a powerful psychological mechanism of influence theory, based on reciprocity, affection, and recognition in nature, trust influences people's behavior and attitude (Acedo-Carmona & Gomila, 2014; Zhu et al., 2013), and brand image will support consumers in recognizing their needs and wants, and trust in brand image promotes purchase intention (H. W. Kim et al., 2012; Lien et al., 2015).

Our study proposed the three types of influencers as the observed variables and reflective indicators of social influence. The study of social influence will be more complex and more difficult. At the same time, our research extends the research scope of trust.

Research Framework and Hypotheses Development

Based on the above-mentioned social influence literature, we recognize the correlation between social influence, brand purchase intention, brand image, and trust. Furthermore, we frame the following hypotheses to examine the empirical considerations.

Social Influence on Brand Purchase Intention

Social influence includes a range of distinct but related processes, for example, conformity, peer pressure, persuasion, socialization, and social regulation (Cialdini & Goldstein, 2004). The influence effect and degree of interaction between individuals are restricted by the relationship between communicator and receiver (Kelman, 1958), the status of communicator, personality charm and information credibility, communication skills and

subjective state of receiver, etc. The strength of social influence depends on the relationships among individuals, timing, characteristics of networks, groups, individuals, and viral networking (Domingos & Richardson, 2001). Consumers' purchase intentions can be influenced by celebrity endorsements, advertising and expert recommendations in e-communities (Tang & Yang, 2012).

In summary, we investigate individual decisions from the circles of face-to-face friendship, e-WOM readership, and social media viewership and how those processes and actions create the social influence to impact consumers' brand purchase intentions.

Thus, our first hypothesis is extended to brand purchase intention as follows:

Hypothesis 1: Social influence positively affects brand purchase intention.

Trust and Social Influence

Trust is the core of promoting positive interpersonal relationships in any environment and the center of interaction between people (Berscheid, 1994). It is a powerful psychological mechanism that affects what people do and choose to do (Chin et al., 2009; Dasgupta, 1988). Rousseau et al. (1998) defined trust as a psychological state, referring to the individual intention to accept the failure of expectations based on positive expectations of the behaviors or preferences of others. Therefore, trust includes affective and cognitive trust (Zhu et al., 2013). The former is based on socio-emotional bonds and concentrates on the exchange of socio-emotional benefits between friends (McAllister, 1995), while the latter is based on the evaluation of the other's distinctive personal characteristics such as integrity, ability, credibility, and reliability, to follow the other's suggestions (Zhu et al., 2013). Thus, trust is the antecedent influencing consumer behavior (Chin et al., 2009).

Trust transfer theory states that an individual's trust can be transferred from a trusted source to an unknown target. Thus, an individual willing to follow the recommendations from their best friends, the credibility celebrities, or e-WOM, in turn boosting a positive attitude to the product (Leader et al., 2021; Burger et al., 2001; Davoudi & Chatterjee, 2018; Erdogan, 1999; M. Kim & Kim, 2020).

Therefore, we assume that:

Hypothesis 2: Trust affects (positively) social influence.

Brand Image and Consumer Behavior

Brand image is the general impression consumers perceived, and it can cause personal association and recall

of related series of products (Y. H. Wang & Tsai, 2014). A brand is a name, term, symbol, design, or all of the above, which is used to distinguish personal products and services from competitors (Kotler, 2000). Therefore, brand image does not exist in features, technology, or actual products themselves, but is brought by advertising, promotion or users (Y. H. Wang & Tsai, 2014). Keller (1993) noted that brand image refers to a series of associations related to brand in consumers' memory (Keller, 1993, p. 2; X. Bian & Moutinho, 2011). It is an external clue for consumers to evaluate goods before purchasing in relation to a brand's overall quality superiority relative to other brands (Y. H. Wang & Tsai, 2014).

Research shows that brand image makes customers play a moderating role in the credibility of product quality, and affects their purchase intention and loyalty (X. Bian & Moutinho, 2011). Kahn and Louie (1990) noted that a good brand image has a positive impact on quality and value of products, and has a positive effect on purchase intention. Therefore, we assume that:

Hypothesis 3: Brand image impact on brand purchase intention.

Based on the above theoretical framework, we explore the research model is shown in Figure 1.

Methology

Instrument Development

Measurement items were adapted from previously validated studies and were modified to fix the context of the current study. Social media viewership (three items) was adapted from Lee and Watkins (2016). Face-to-face friendship was measured by three-item scale adapted from X. Wang et al. (2012). Three items measuring eWOM viewership were derived from Putri Utami et al. (2020), Sa'ait et al. (2016), and Singh and Matsui (2017), and social influence (three items) measurement items were adapted from Q. Bian and Forsythe (2012). Articles about trust (three items) were drawn from Zeng et al. (2020). The items measuring brand image (three items) came from Nyadzayo and Khajehzadeh (2016). Three measurement variables of brand purchase intentions were adapted from Park et al. (2021) and Hsu and Lin (2016). All items were measured using a five-point Likert scale (1–5), where 1 = strongly disagree and 5 = strongly agree.

Since the data were collected in China, the questionnaire was translated into Chinese by bilingual professionals and reviewed by other professionals to ensure that the translation was not distorted. Furthermore, we also conducted the pilot test with 30 samples to improve content

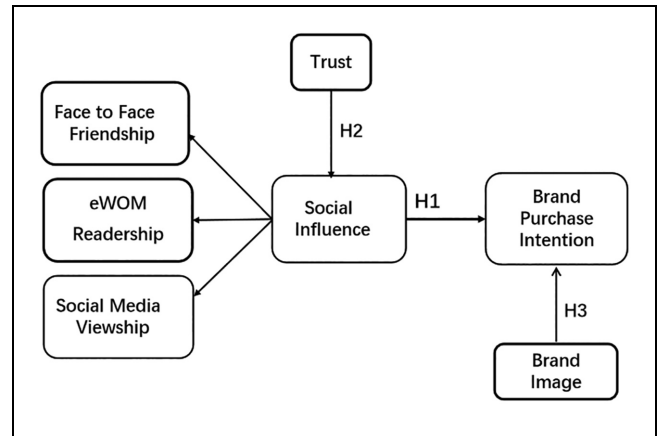


Figure 1. Research model.

validity. This pre-test resulted in a slight modification of the questionnaire, which is shown in Table 1.

Data Collection and Sample Characteristics

We conducted an empirical survey in China to test the proposed research model. Due to the ubiquity of the Internet and various APPs such as live broadcasts, Weibo, shopping and mobile payment, etc., Chinese people are highly dependent on mobile phones, which is very suitable for our research. We distribute questionnaires through Sojump. To ensure the validity and authenticity of the questionnaire, we set up a mobile phone questionnaire and paid for it. So far, 402 questionnaires have been recovered from June 2021 to July 2021, and 398 were valid questionnaires. The ratio of the number of replies to the number of items is 10:1, which meets the requirements of the questionnaire survey.

The demographic characteristics of respondents are shown in detail in Table 2. Interestingly, more than half of the respondents followed at least one celebrity, and all have online brand-buying experience. As for age, 206 respondents were male (52.02%), 190 were female (47.98%), 288 (72.73%) were aged between 21 and 30, and 108 (27.27%) were between 31 and 40. Moreover, 76.77% of participants said they buy brand goods once a year and spend at least 10,000 RMB on brand products. The questionnaire content is suitable for this study based on the demographic information provided.,

Data Analysis and Results

The current study used the AMOS 23.0 program to analyze the collected data. The analytical procedure was a two-step process: the measurement and the structural model were checked by structural equation modeling (SEM) methods and exploratory factor analysis (EFA).

Table 1. The Measure.

Construct	Items	Adapted from
Face-to-face friendship	I seek out the advice of my friends and family before I choose a product. I trust the products recommended by my friends for friendships I have with my peers mean a lot to me.	X. Wang et al. (2012)
Social media view-ship	I asked my peers for information about the product. The knowledge of brands comes from online celebrities. Live streaming of online celebrities will guide the purchase of brand goods. The celebrity is familiar with the brand, so I pay attention to the brand she recommended.	Lee and Watkins (2016)
e-WOM readership	An accurate and quality message about an online product is trustworthy and helpful. The probability of choosing this brand because the online review has sufficient reasons supporting the opinions. The relevance, timeliness, and comprehensiveness of online reviews affect the reliability of the information.	Singh and Matsui (2017), Putri Utami et al. (2020), Sa'ait et al. (2016)
Trust	Source credibility determines the reliability of network information. Brand live streaming can be trusted, since they keep the promises they make to us.	Kenworthy et al. (2016)
Brand image	The network's evaluation of the brand is relatively objective. The brand I purchased provides good value for the money. The brand I purchased has personality and is interesting. The brand is different from competing brands.	Nyadzayo and Khajezadeh (2016)
Social influence	Having certain brands is a way to express oneself. Owning brand goods attracts attention. Brand goods are important to me because they make me feel accepted in my social circle.	Q. Bian and Forsythe (2012)
Brand purchase intention	I am highly likely to purchase brand items. I would recommend brand goods to other people. I'm likely to purchase brand items.	Park et al. (2021), Hsu and Lin (2016)

Measurement Model

This study applied the threshold of 0.6 as the standard value of factor loading (Hair et al., 2012). The analysis results show that the factor loading of all 15 items in the measurement model exceeded the 0.6 criteria, and all of them were retained for reliability and validity evaluation.

According to Nunnally (1978), the composite reliability (CR) values should be higher than 0.7. The results showed that the combined reliability of the five constructs ranged from 0.753 to 0.862 and exceeded the recommended criteria, achieving acceptable reliability (Table 3). The average variance extracted (AVE) for each construct is higher than 0.5 (Fornell & Larcker, 1981). Table 3 shows that the AVE values of the five constructs ranged from 0.505 to 0.677 and exceeded the threshold of 0.5. Furthermore, all constructs met the conditions suggested by Fornell and Larcker (1981), indicating that the instrument achieves convergent validity.

Additionally, the measurement model achieves discriminant validity (Fornell & Larcker, 1981) when the squared root of the AVE for each construct exceeds the correlation between the shared correlations between each

pair of constructs as seen in Table 4. This result confirmed the discriminant validity of the measurements.

Common Method Variance and Pearson Test

The contents of the questionnaire may cause co-variability due to various factors, so it is necessary to use statistical methods to check the issue of common method bias (CMB). SPSS was used to perform this test as factor analysis, in which all the items were loaded with a threshold to attain one factor. The results showed that a single element contributes to 39% of the total variance extracted, which is well below 50%. This confirms that CMB is not a significant concern for this study (Hew & Syed Abdul Kadir, 2016).

Pearson correlation coefficients were used to verify items' multicollinearity in our study. Correlation between the items lower than .7 indicated the good multicollinearity of the items (Dancey, 2004). The results support the hypothesis because the correlation coefficient between social influence and brand image is .56, which is the required range (Dancey, 2004). Therefore, the results

Table 2. Respondents' Demographic Characteristics.

Measure	Items	Frequency	Percentage
Gender	Male	206	52.02
	Female	190	47.98
Age	21–30	288	72.73
	31–40	108	27.27
Education	Junior high	16	4.04
	High school	129	32.58
	Bachelor's degree	139	35.10
	Master's degree or above	112	28.28
Where do you usually purchase brand products?	Online shopping	96	24.24
	Live streaming shopping	105	26.52
	Physical stores shopping	88	22.22
	Overseas shopping	107	27.02
Frequency of watching live streaming	Once a week	286	72.22
	Almost everyday	110	27.78
Number of online influencers followed	One person	196	49.49
	Two	86	21.97
	Three	31	8.33
	Four	9	2.27
	Five	9	2.27
	>Five	62	15.66
Period of time following online influencers	1 year	240	60.61
	2 years	110	27.78
	3 years	18	4.55
	>3 years	28	7.07
Frequency of online brand purchasing	One time	300	77.02
	Two times	20	6.31
	Three times	6	2.78
	>Three times	50	13.89
The amount of brand goods purchased online (RMB)	Less than 10,000	300	76.77
	10,000–30,000	20	6.06
	30,001–50,000	6	2.53
	50,001–100,000	25	7.32
	100,000–200,000	25	7.32
	>200,000	Zero	Zero

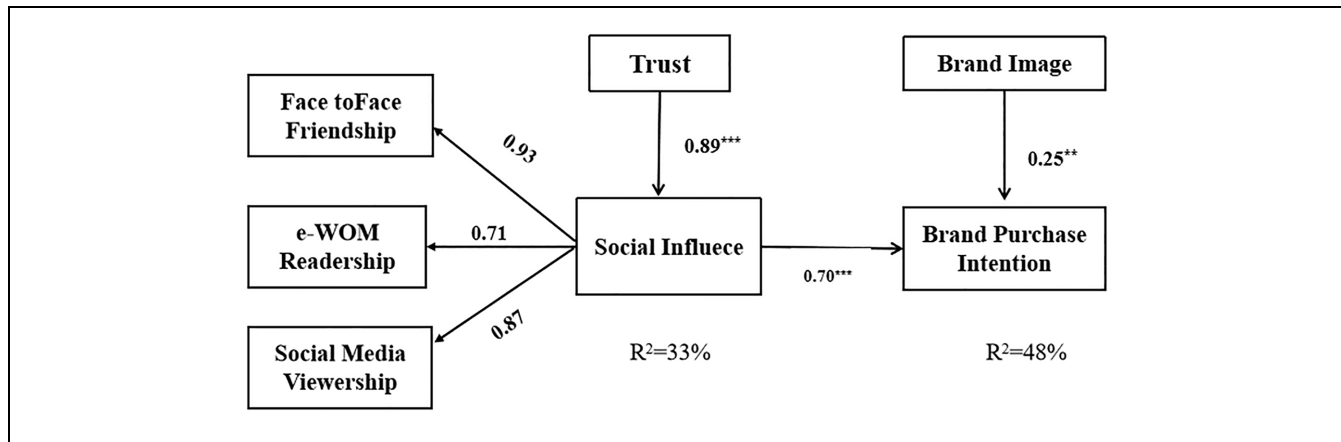
Table 3. Factor Loadings, CR and AVE Values.

Construct	Items	Factor loading	CR	AVE
Social influence	SI1	0.709	0.776	0.536
	SI2	0.768		
	SI3	0.717		
Face to face friendship	FTF1	0.676	0.753	0.505
	FTF2	0.794		
	FTF3	0.655		
E-WOM readership	e-WOM1	0.720	0.813	0.593
	e-WOM2	0.831		
	e-WOM3	0.754		
Social media viewership	SMV1	0.867	0.862	0.677
	SMV2	0.763		
	SMV3	0.835		
Brand purchase intention	PI1	0.737	0.803	0.583
	PI2	0.911		
	PI3	0.612		
Trust	TR1	0.766	0.784	0.549
	TR2	0.634		
	TR3	0.812		
Brand image	BI1	0.711	0.826	0.614
	BI2	0.783		
	BI3	0.851		

Table 4. Results of Discriminant Validity.

	AVE	BI	PI	SI	TR
BI	0.614	0.784			
PI	0.583	0.700	0.764		
SI	0.536	0.595	0.629	0.732	
TR	0.549	0.765	0.672	0.594	0.741

Note. SI = social influence, PI = brand purchase intention; TR = trust; BI = brand image.

**Figure 2.** The path coefficients of the structural model.**Table 5.** Testing Results of the Hypotheses.

Hypothesis		Path coefficient	Result
H1	SI → PI	0.70***	Supported
H2	TR → SI	0.89***	Supported
H3	BI → PT	0.25	Supported

Note. PI = brand purchase intention; SI = social influence; TR = trust.

*** $p < .05$; all path coefficients are in standardized format.

indicate that multicollinearity was not a severe issue in this study.

Structural Equation Model

The structural model and hypothesis were tested by applying SEM via the AMOS 23.0 software. The result illustrates overall fit indicators of the structural model, in which χ^2/df (2.429), CFI (0.941), NFI (0.904), GFI (0.901), IFI (0.941), TLI (0.932), RMSEA (0.060), and RMR (0.023) all achieved the cut-off criteria. Therefore, the structural model satisfied the fit indices. Figure 2 also illustrates the hypotheses testing results in terms of the principal path coefficients and explains endogenous variables' variance (R^2). The results indicated that all paths of the research model were supported. In detail, the relation between social influence and brand purchase intention was statistically significant ($\beta = .72$; $p < .001$), thus

supporting H1. As expected, the effect of trust and social influence ($\beta = .93$, $p < .001$), and brand image is positively associated with brand purchase intention ($\beta = .27$, $p < .005$), supporting H2 and H3. Additionally, the structural model accounted for 33% and 48% of the variances in social influence and brand purchase intention, respectively. The results of the hypotheses testing are depicted in Table 5, and the statistical analysis of the study show in Figure 3.

Reflective Specification of Social Influence Test

Figures 4 and 5 show that the results for both the first-order and second-order reflection model are statistically significant.

The results of the second-order reflective model indicate that social influence has a substantial impact on face-to-face friendship ($\beta = .851$, $p < .001$), e-WOM

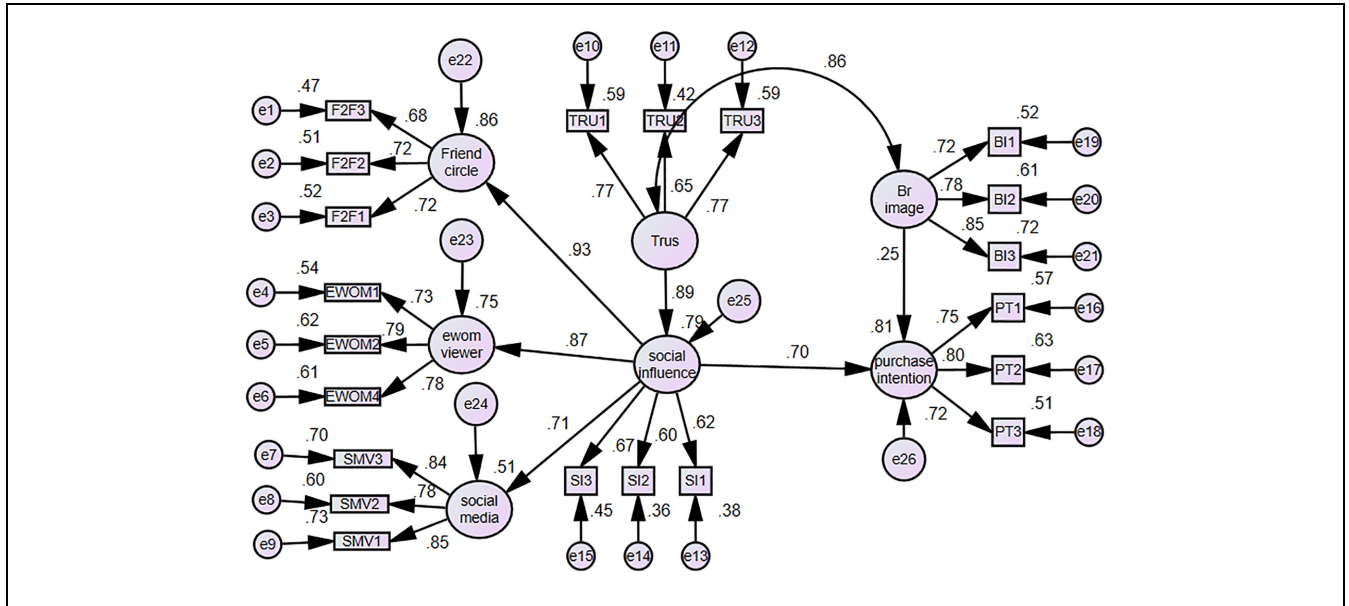


Figure 3. The statistic analysis.

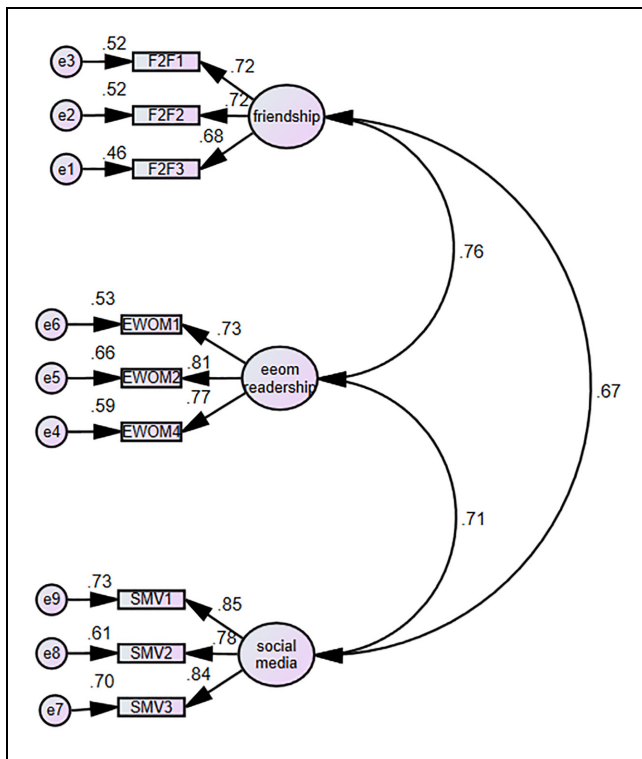


Figure 4. First-order model.

readership ($\beta = .896, p < .001$) and social media viewership ($\beta = .791, p < .001$) as well as a good fit: $\chi^2 = 54.640$, $DF = 24$, $p = .000$, $\chi^2/df = 2.277$, $GFI = 0.971$, $AGFI = 0.946$, $CFI = 0.981$, $IFI = 0.981$, $TLI = 0.972$, $RMSEA = 0.057$, and $RMR = 0.017$. The

results indicated that the three response indicators of social influence have good validity and reliability, which meet the standards of the reflective model.

Conclusions and Implications

This study explores the research on social influence impact on product purchase intention and brand image on brand purchase intention. The research results align with Lien et al. (2015) and Aghekyan-Simonian et al. (2012). That is, social influence has a positive effect on purchase intention; The impact of purchase intention has a significant correlation.

With the development of networks and technology, the sources of information on social influence have become diverse and complex. Social platforms and social media make individuals get the message about products easier than traditional advertising models. In this case, the study of the influence of trust on social influence in psychological mechanisms is critical. Our research explores the relationship between the two and fills in the research gap. In addition, the network influencers enhance consumers' cognition, attitude, and decision-making on products. Among the various influencers, which influence is the most convincing is particularly important for the formulation of marketing strategies of enterprises and organizations. Our research fills the corresponding research gap again.

As the reflective dimension of social influence, our research shows that information from face-to-face friendships is the most influential; close friends are the ones who understand each other's psychological demands and

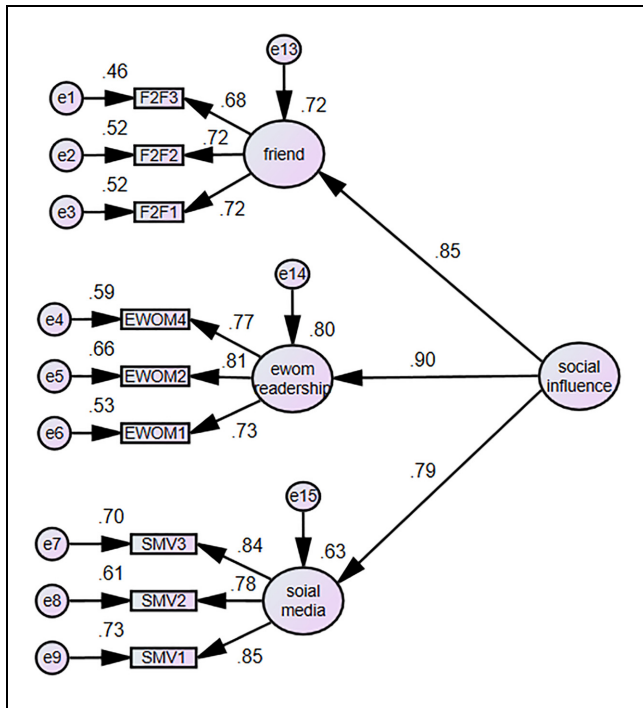


Figure 5. Second- order model.

reactions best; the second is the network influence in live-streaming. It is a more powerful influence than that from e-WOM viewership. The live-streaming mode can enhance the interaction between people and promote emotions, allow consumers to see the product's relevant information intuitively, and increase the product's awareness and trust.

Therefore, our research provides a basis for enterprises to formulate and implement marketing strategies.

Theoretical Implications

The relationship between social influence and consumer psychological mechanism trust is relatively complex. Consumer behavior change cannot be explained by a single mechanism, but reflects the process of multiple potential causal relationships. Moreover, the causal relationship between social influence and behavioral change is partly manifested not as a primary influence factor but as an interaction, exhibiting a dynamic relationship related to contextual changes provided by technological development. The research on social influence under socio-technical changes which have important implications for both scholars and practitioners. The diversity of factors influencing consumer buying behavior highlights the need for different approaches to exploring models, as well as a more complex psychological mechanism in research model.

Our study provides empirical evidence on the extent to which individuals influence changes in consumer behavior. These have been less studied in the previous literature. The influence of face-to-face friendship is the most powerful. Therefore, marketing strategies must be carefully tuned to different marketing environments and maximize their impact on influencers.

First, this study verifies that friendship is the most trustworthy and reliable among all interpersonal and social relationships. It also confirms the conclusion of Mead et al. (2014) and McMillan et al. (2018) that people constantly change their behaviors by viewing their friends' opinions to maintain a friendship or ensure belonging to the friends' group. In the study by M. Kim and Kim (2020), friendship significantly impacts on individual personal life satisfaction and mental health (De Vries et al., 2018). Therefore, this may be the core value of people's trust in friends, along with the reason that generates compliance with the change of thinking when friends recommend a product. The intimate relationship between circle friends is the core and the strongest association, which is suitable for viral marketing. Our research lays the theoretical foundation for viral marketing and enables customers' friends to become corporate customers through customer recommendations.

Secondly, based on the theory of cognitive trust, celebrities' credibility and reliability are the key factors that influence consumers' attitude toward products. The psychological mechanism of internalization influence should come from consumers' demand for improving their image or perceiving that the celebrity represents their ideal image. As M. Kim and Kim (2020) suggested, consistency with the inner ideal self-image is the social influence of internalization motivation. At the same time, this study fills in the theoretical research on the impact of celebrity congruence with the consumer in consumers' purchase behavior, which can serve as the theoretical basis for similar research and expand academic research, which is a contribution of this study.

Third, faced with unfamiliar products, people often refer to the information on the Internet. Demonstrating the reliability and credibility of information sources is the key to selling products online.

We emphasize the intermediary role of social influence between reference sources and purchase decisions. This kind of emphasis has been limited in previous studies. Social relations directly or indirectly affect people's purchase behavior, showing varying psychological changes in compliance with recommendation, identification, and internalization. Trust is the psychological mechanism that influence consumers' willingness and decision. Marketers need to adjust marketing strategies according to the theory.

Practical Implications

The extant literature shows that social influences affect consumers' attitudes toward a product brand.

- (1) Marketing managers focus on the power of customer circles. Marketing activities should establish friend-like relationship with customers while product value matches customers' needs; Encourage social shopping with referral links, shared coupons, and money-saving offers.
- (2) Select the appropriate network influencers and select a celebrity who is attractive and similar to the target customer group as the brand spokesperson.
- (3) Based on focusing on brand management, managing customer reviews, handling customer experience with a positive attitude, and increasing the credibility of positive review content.

Limitations

The subjects of the experiments come from different classes, and their brand consumption habits and concepts are pretty other. People usually need to refer to various sources of credible information before making brand shopping decisions. We just made a simple comparison of a few variables.

Future research will explore how companies build friendship with consumers and discuss the influence of friendship on marketing performance compared with the previous relationship with consumers.

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Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

References

- Acedo-Carmona, C., & Gomila, A. (2014). Personal trust increases cooperation beyond general trust. *PLoS One*, 9(8), e105559.
- Aghekyan-Simonian, M., Forsythe, S., Suk Kwon, W., & Chat-taraman, V. (2012). The role of product brand image and online store image on perceived risks and online purchase intentions for apparel. *Journal of Retailing and Consumer Services*, 19(3), 325–331.
- Al-Adwan, A. S., Alrousan, M. K., Yaseen, H., Alkufahy, A. M., & Alsoud, M. (2022). Boosting online purchase intention in high-uncertainty-avoidance societies: A signaling theory approach. *Journal of Open Innovation Technology Market and Complexity*, 8(3), 136. <https://doi.org/10.3390/joitmc8030136>
- Al Rabadi, S. (2022). Cristiano Ronaldo, Football and Sustainable Governance. <https://www.researchgate.net/publication/366095773>
- Alsoud, M., Al-Muani, L., & Alkhazali, Z. (2022). Digital platform interactivity and Jordanian social commerce purchase intention. *International Journal of Network Science*, 6(2), 285–294.
- Argo, J. J., & Dahl, D. W. (2020). Social influence in the retail context: A contemporary review of the literature. *Journal of Retailing*, 96(1), 25–39.
- Ayeh, J. K., Au, N., & Law, R. (2013). "Do we believe in tripadvisor?" Examining credibility perceptions and online travelers' attitude toward using user-generated content. *Journal of Travel Research*, 52(4), 437–452.
- Bartal, A., Pliskin, N., & Ravid, G. (2019). Modeling influence on posting engagement in online social networks: Beyond neighborhood effects. *Social Networks*, 59(10), 61–76.
- Berscheid, E. (1994). Interpersonal relationships. *Annual Review of Psychology*, 45, 79–129.
- Bian, Q., & Forsythe, S. (2012). Purchase intention for luxury brands: A cross cultural comparison. *Journal of Business Research*, 65(10), 1443–1451.
- Bian, X., & Moutinho, L. (2011). The role of brand image, product involvement, and knowledge in explaining consumer purchase behaviour of counterfeits: direct and indirect effects. *European Journal of Marketing*, 45(1/2), 191–216.
- Burger, J. M., Soroka, S., Gonzago, K., Murphy, E., & Somervell, E. (2001). The effect of fleeting attraction on compliance to requests. *Personality and Social Psychology Bulletin*, 27, 1578–1586.
- Cambra-Fierro, J., Gao, L. (X.), & Melero-Polo, I. (2021). The power of social influence and customer-firm interactions in predicting non-transactional behaviors, immediate customer profitability, and long-term customer value. *Journal of Business Research*, 125, 103–119.
- Chih, W.-H., Wang, K.-Y., Hsu, L.-C., & Huang, S.-C. (2013). Investigating electronic word-of-mouth effects on online discussion forums: The role of perceived positive electronic word-of-mouth review credibility. *Cyberpsychology Behavior and Social Networking*, 16(9), 658–668. <https://doi.org/10.1089/cyber.2012.0364>
- Chin, A. J., Wafa, S., & Ooi, A. Y. (2009). The effect of internet trust and social influence towards willingness to purchase online in Labuan, Malaysia. *International Business Research*, 2, 72–81.
- Choi, S. M., & Rifon, N. J. (2012). It is a match: The impact of congruence between celebrity image and consumer ideal self on endorsement effectiveness. *Psychology and Marketing*, 29(9), 639–650.
- Cialdini, R. B., & Goldstein, N. J. (2004). Social influence: Compliance and conformity. *Annual Review of Psychology*, 55(1), 591–621.

- Dancey, J., R., (2004). *Statistics without maths for psychology: Using SPSS for Windows*. Prentice H.
- Dasgupta, P. (1988). Trust as a commodity. In D. Gambetta (Ed.), *Trust: Making and breaking cooperative relations* (pp. 49–72). B. Blackwell.
- Davoudi, A., & Chatterjee, M. (2018). Social trust model for rating prediction in recommender systems: Effects of similarity, centrality, and social ties. *Online Social Networks and Media*, 7, 1–11.
- De Vries, E. L. E., Fennis, B. M., Bijmolt, T. H. A., Ter Horst, G. J., & Marsman, J. B. C. (2018). Friends with benefits: Behavioral and fMRI studies on the effect of friendship reminders on self-control for compulsive and non-compulsive buyers. *International Journal of Research in Marketing*, 35(2), 336–358.
- Deutsch, M., & Gerard, H. B. (1955). A study of normative and informational social influences upon individual judgment. *The Journal of Abnormal and Social Psychology*, 51(3), 629.
- Dimitrieska, S., & Efremova, T. (2021). The effectiveness of the influencer marketing. *SPE Economics & Management*, 18(1), 109–118.
- Doh, S. J., & Hwang, J. S. (2009). How consumers evaluate eWOM (electronic word-of-mouth) messages. *CyberPsychology & Behavior*, 12, 193–197.
- Domingos, P., & Richardson, M. (2001). *Mining the network value of customers* [Conference session]. Proceedings of the 7th ACM Conference on Knowledge Discovery and Data Mining, New York, USA.
- Doney, P. M., & Cannon, J. P. (1997). An examination of the nature of trust in buyer-seller relationships. *Journal of Marketing*, 61(2), 35–51.
- Engels, R. C., Vitaro, F., Blokland, E. D., de Kemp, R., & Scholte, R. H. (2004). Influence and selection processes in friendships and adolescent smoking behaviour: The role of parental smoking. *Journal of Adolescence*, 27, 531–544.
- Erdogan, B. Z. (1999). Celebrity endorsement: A literature review. *Journal of Marketing Management*, 15(4), 291–314.
- Filieri, R., McLeay, F., Tsui, B., & Lin, Z. (2018). Consumer perceptions of information helpfulness and determinants of purchase intention in online consumer reviews of services. *Information Management*, 55(8), 956–970.
- Fleury, S., Agnès, A., Cados, L., Denis-Lutard, Q., Duchêne, C., Rigaud, N., & Richir, S. (2020). Effects of social influence on idea selection in creativity workshops. *Thinking Skills and Creativity*, 37, 100691.
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–346.
- Gellerstedt, M., & Arvemo, T. (2019). The impact of word of mouth when booking a hotel: Could a good friend's opinion outweigh the online majority? *Information Technology & Tourism*, 21, 289–311. <https://doi.org/10.1007/s40558-019-00143-4>
- Godey, B., Manthiou, A., Pederzoli, D., Rokka, J., Aiello, G., Donvito, R., & Singh, R. (2016). Social media marketing efforts of luxury brands: Influence on brand equity and consumer behavior. *Journal of Business Research*, 69(12), 5833–5841.
- Hair, J. F., Sarstedt, M., Ringle, C. M., & Mena, J. A. (2012). An assessment of the use of partial least squares structural equation modeling in marketing research. *Journal of the Academy of Marketing Science*, 40(3), 414–433.
- Hew, T. S., & Syed Abdul Kadir, S. L. (2016). Understanding cloud-based VLE from the SDT and CET perspectives: Development and validation of a measurement instrument. *Computers & Education*, 101, 132–149.
- Hsu, C. L., & Lin, J. C. C. (2016). Effect of perceived value and social influences on mobile app stickiness and in-app purchase intention. *Technological Forecasting and Social Change*, 108, 42–53.
- Huhn, R., Brantes Ferreira, J., Sabino de Freitas, A., & Leão, F. (2018). The effects of social media opinion leaders' recommendations on followers' intention to buy. *Review of Business Management*, 20(1), 57–73.
- Hui, T. X. (2017). The effect of source credibility on consumers' purchase intention in Malaysia online. *J. Arts Soc. Sci*, 1(1), 12–20.
- Kahn, B. E., & Louie, T. A. (1990). Effects of retraction of price promotions on brand choice behavior for variety-seeking and last-purchase-loyal consumers. *Journal of Marketing Research*, 27(3), 279–289.
- Kelman, H. C. (1958). Compliance, identification, and internalization: Three processes of attitude change. *Journal of Conflict Resolution*, 2(1), 51–60.
- Kenworthy, J. B., Voci, A., Ramiah, A. A., Tausch, N., Hughes, J., & Hewstone, M. (2016). Building trust in a post-conflict society: An integrative model of cross-group friendship and intergroup emotions. *Journal of Conflict Resolution*, 60(6), 1041–1070.
- Kim, H. W., Xu, Y., & Gupta, S. (2012). Which is more important in Internet shopping, perceived price or trust? *Electronic Commerce Research and Applications*, 11(3), 241–252.
- Kim, M., & Kim, J. (2020). How does a celebrity make fans happy? Interaction between celebrities and fans in the social media context. *Computers in Human Behavior*, 111, 106419.
- Kim, M., & Qu, H. (2017). The moderating effects of three reference groups on Asian pleasure travelers' destination value. *Journal of Travel & Tourism Marketing*, 34(7), 892–904.
- Kirschman, L. (2022). <https://www.washington.edu/news/2022/10/19/these-factors-have-the-biggest-impact-on-influencer-marketing-effectiveness/>
- Kotler, P. (Ed.). (2000). *Marketing management* (10th ed.). Prentice-Hall.
- Kucharska, W. (2018). *Personal brand value and social media. The top football players' case* [Conference session]. Innovation Management and Education Excellence Through Vision 2020, International Business Information Management Association (IBIMA), Milan, Italy.
- Leader, A. E., Burke-Garcia, A., Massey, P. M., & Roark, J. B. (2021). Understanding the messages and motivation of vaccine hesitant or refusing social media influencers. *Vaccine*, 39(2), 350–356.
- Lee, J. E., & Watkins, B. (2016). YouTube vloggers' influence on consumer luxury brand perceptions and intentions. *Journal of Business Research*, 69(12), 5753–5760.
- Lien, C. H., Wen, M. J., Huang, L. C., & Wu, K. L. (2015). Online hotel booking: The effects of brand image, price, trust and value on purchase intentions. *Asia Pacific Management Review*, 20(4), 210–218.

- Lin, J., Luo, Z., Cheng, X., & Li, L. (2019). Understanding the interplay of social commerce affordances and swift guanxi: An empirical study. *Information Management*, 56, 213–224.
- Loughran Dommer, S., Swaminathan, V., & Ahluwalia, R. (2013). Using differentiated brands to deflect exclusion and protect inclusion: The moderating role of self-esteem on attachment to differentiated brands. *Journal of Consumer Research*, 40(4), 657–675.
- Majali, T., Alsoud, M., Yaseen, H., Almajali, R., & Barkat, S. (2022). The effect of digital review credibility on Jordanian online purchase intention. *International Journal of Network Science*, 6(3), 973–982.
- McAllister, D. J. (1995). Affect- and cognition-based trust as foundations for interpersonal cooperation in organizations. *Academy of Management Journal*, 38, 24–59.
- McMillan, C., Felmlee, D., & Osgood, D. W. (2018). Peer influence, friend selection, and gender: How Network processes shape adolescent smoking, drinking, and Delinquency. *Social Networks*, 55, 86–96.
- Mead, E. L., Rimal, R. N., Ferrence, R., & Cohen, J. E. (2014). Understanding the sources of normative influence on behavior: The example of tobacco. *Social Science & Medicine*, 115, 139–143.
- Meng, L. (.), Duan, S., Zhao, Y., Lü, K., & Chen, S. (2021). The impact of online celebrity in livestreaming E-commerce on purchase intention from the perspective of emotional contagion. *Journal of Retailing and Consumer Services*, 63, 102733. <https://doi.org/10.1016/j.jretconser.2021.102733>
- Mi, L., Zhu, H., Yang, J., Gan, X., Xu, T., Qiao, L., & Liu, Q. (2019). A new perspective to promote low-carbon consumption: The influence of reference groups. *Ecological Economics*, 161, 100–108.
- Nunnally, J. (Ed.). (1978). Fundamentals of factor analysis. In *Psychometric theory* (2nd ed., pp. 327–404) McGraw-Hill.
- Nyadzayo, M. W., & Khajehzadeh, S. (2016). The antecedents of customer loyalty: A moderated mediation model of customer relationship management quality and brand image. *Journal of Retailing and Consumer Services*, 30, 262–270.
- Park, J., Hyun, H., & Thavisay, T. (2021). A study of antecedents and outcomes of social media WOM towards luxury brand purchase intention. *Journal of Retailing and Consumer Services*, 58, 102272.
- Perry, B. L., & Ciciurkaite, G. (2019). Contributions of personality to social influence: Contingent associations between social network body size composition and BMI. *Social Science & Medicine*, 224(24), 1–10.
- Putri Utami, S., Setyowati, N., & Mandasari, P. (2020). Celebrity brand ambassador and e-WOM as determinants of purchase intention: A survey of Indonesian celebrity cake. *E3S Web of Conferences*, 142, 1–9.
- Rahal, A. (2020, January 10). Is influencer marketing worth it in 2020? *Forbes*. <https://www.forbes.com/sites/theyec/2020/01/10/is-influencer-marketing-worth-it-in-2020/>
- Rousseau, D. M., Sitkin, S. B., Burt, R. S., & Camerer, C. (1998). 'Not so different after all: A cross discipline view of trust'. *Academy of Management Review*, 23, 393–404.
- Sa'ait, N., Kanyan, A., & Nazrin, M. F. (2016). The effect of e-WOM on customer purchase intention. *International Academic Research Journal of Social Science*, 2(1), 73–80.
- Schwarzwald, J., Koslowsky, M., & Agassi, V. (2001). Captain's leadership type and police officers' compliance to power bases. *European Journal of Work and Organizational Psychology*, 10(3), 273–290.
- Simons-Morton, B. G., & Farhat, T. (2010). Recent findings on peer group influences on adolescent smoking. *Journal of Primary Prevention*, 31(4), 191–208.
- Singh, M., & Matsui, Y. (2017). How long tail and trust affect online shopping behavior: An extension to UTAUT2 framework. *Pacific Asia Journal of the Association for Information Systems*, 9(4), 1–24.
- Statista. (2021). *Online reviews - Statistics & Facts*. Retrieved June 8, 2021, from <https://n9.cl/8jlcl>
- Tang, X., & Yang, C. C. (2012). Ranking user influence in healthcare social media. *ACM Transactions on Intelligent Systems and Technology*, 3(4), 1–21.
- Tobon, S., & García-Madariaga, J. (2021). Influencers vs the power of the crowd: A research about social influence on digital era. *Estudios Gerenciales*, 37(161), 601–609. <https://doi.org/10.18046/j.estger.2021.161.4498>
- Tucker, J. S., de la Haye, K., Kennedy, D. P., Green, H. D., Jr, & Pollard, M. S. (2014). Peer influence on marijuana use in different types of friendships. *Journal of Adolescent Health*, 54, 67–73.
- Turner, J. C., & Oakes, P. J. (1986). The significance of the social identity concept for social psychology with reference to individualism, interactionism and social influence. *British Journal of Social Psychology*, 25(3), 237–252.
- Wang, X., Yu, C., & Wei, Y. (2012). Social media peer communication and impacts on purchase intentions: A consumer socialization framework. *Journal of Interactive Marketing*, 26(4), 198–208.
- Wang, Y. H., & Tsai, C. F. (2014). The relationship between brand image and purchase intention: Evidence from award winning mutual funds. *The International Journal of Business and Finance Research*, 8(2), 27–40.
- Yang, J., He, X., & Lee, H. (2007). Social reference group influence on mobile phone purchasing behaviour: A cross-nation comparative study. *International Journal of Mobile Communications*, 5(3), 319–338.
- Zeng, P., Zhao, X., Xie, X., Long, J., Jiang, Q., Wang, Y., Qi, L., Lei, L., & Wang, P. (2020). Moral perfectionism and online prosocial behavior: The mediating role of moral identity and the moderating role of online interpersonal trust. *Personality and Individual Differences*, 162, 110017.
- Zhang, T., Tao, D., Qu, X., Zhang, X., Zeng, J., Zhu, H., Zhu, H., & Zhuc, H. A. N. (2020). Automated vehicle acceptance in China: Social influence and initial trust are key determinants. *Transportation Research*, 112, 220–233.
- Zhuang, Y. B., Li, Z. H., & Zhuang, Y. J. (2021). Identification of influencers in online social networks: Measuring influence considering multidimensional factors exploration. *Heliyon*, 7, e06472.
- Zhu, W., Newman, A., Miao, Q., & Hooke, A. (2013). Revisiting the mediating role of trust in transformational leadership effects: Do different types of trust make a difference? *Leadership Quarterly*, 24(1), 94–105.